

SAFETY FIRST – Before You Start:

Item	Reason
Wear gloves, goggles, and a mask	Prevent contact with skin, eyes, or inhalation of dust or fumes
Work in a well-ventilated area	Avoid buildup of chlorine fumes
Use plastic or non-corrosive containers	Calcium hypochlorite reacts with metal and acids
Always add chemical to water , never the reverse	Prevents violent reactions or splashing

Advantages in Water Treatment:

Advantage	Explanation
High Disinfection Power	Delivers 65–70% available chlorine, effectively kills bacteria, viruses, protozoa, and algae.
Portable and Easy to Store	As a stable solid, it's ideal for remote or rural water treatment systems without refrigeration.
Long Shelf Life	Doesn't degrade quickly like liquid chlorine; maintains potency if stored dry.
Quick Acting	Acts rapidly to disinfect contaminated water — useful for emergency or disaster relief settings.
Cost-Effective at Scale	Powder form is cheaper to transport and dose than liquid equivalents for large-scale municipal use.
Residual Protection	Leaves residual chlorine in water to protect against recontamination during distribution.
Simple Dosing	Can be measured and added easily with basic equipment or manually in low-tech setups.

Disadvantages in Water Treatment:

Disadvantage	Explanation
Sensitive to Moisture	Hygroscopic and decomposes on contact with humidity — must be kept very dry.
Chlorine Taste & Odor	Can cause an unpleasant taste and odor in treated drinking water if overdosed.

Disadvantage	Explanation
Dangerous Reactions	Reacts violently with acids and organics — risks include fire, explosion, or toxic gas release.
Sludge Formation	Can leave behind calcium residue or scale in treatment tanks and distribution systems.
Chlorine Demand Variability	Effectiveness depends on pH and organic load — may require careful testing and adjustment.
Handling Safety	Requires trained personnel and PPE; hazardous if spilled or mishandled.
Environmental Concern	Overuse can lead to chlorinated organic compounds forming in water (e.g., trihalomethanes).

Formula:

Required Chlorine Dose (mg/L) × Water Volume (L)

----- = grams of $\text{Ca}(\text{ClO})_2$

(Available Chlorine % × 10)

Example for 3 mg/L target:

$(3 \times 1000) / (70 \times 10) = 4.29$ grams

- 4.3 g for 3 mg/L in 1000 L
- Dissolve in ~10 L clean water
- Slowly mix into the 1000 L tank

Do NOT:

- Use less than **5 liters** for this amount — it may generate heat or fumes
- Use hot water — this accelerates chlorine gas release
- Mix in metal containers — chlorine corrodes metals